

Sahil Pal

Software Engineer

Email | GitHub | LinkedIn | Portfolio | Phone

Summary

Software engineer focused on building production-grade systems and developer tools, with experience in REST APIs, async pipelines, worker queues, and scalable data services.

Technical Skills

Languages: Java, C++, TypeScript, JavaScript, SQL

Backend & Systems: REST APIs, JWT, OAuth, SSE, Worker Queues, Async Pipelines

Databases & Search: PostgreSQL, MongoDB, Redis, pgvector, Full-Text Search

Tools & Frameworks: Vert.x, Express, Next.js, React, Tailwind CSS, Vite, Manifest V3, GitHub Actions

Concepts: RAG, Vector Search, Event-Driven Architecture

Projects

• RAG Doc Analyzer

Live | GitHub

Document Q&A platform — Next.js, Express, PostgreSQL, Gemini

- Implemented a grounded RAG pipeline using Gemini embeddings, pgvector similarity search, and PostgreSQL full-text fallback with SSE response delivery.
- Built a chat-first platform with Firebase authentication, JWT sessions, persistent workspaces, and multi-document conversations.
- Designed async ingestion (Multer, SHA-256 deduplication, Supabase Storage) and a PostgreSQL worker queue using `FOR UPDATE SKIP LOCKED`.
- Added ownership checks, quotas, and health endpoints to ensure reliable production workflows.

• LeetSync

Live | GitHub

Chrome extension + product site — TypeScript, Vite, Manifest V3

- Built a Manifest V3 Chrome extension to capture LeetCode editor content and metadata, converting solutions into structured Markdown.
- Implemented page-context injection (popup + Shadow DOM) for in-tab capture of problem-solving notes and metadata.
- Automated GitHub synchronization via Contents API across structured DSA folders, eliminating manual workflows.
- Launched a companion site with installation guides and setup scripts to streamline onboarding.

Experience

• Intern

June 2025 – July 2025

Kristu Jayanti Software Development Centre — Kristu Jayanti University

- Built a Java 17 + Vert.x backend exposing REST APIs for authentication, profiles, items, search, and moderation.
- Implemented production-grade authentication including JWT sessions, email verification, password reset, Redis token blacklisting, and role-based access control.
- Designed MongoDB-backed workflows with SMTP notifications, scheduled cleanup jobs, and asynchronous handlers coordinating database, cache, and email systems.

Education

• Bachelor of Computer Applications (BCA)

2023 – 2026

Kristu Jayanti College, Bengaluru